

Probability and Statistics Final Exam Solution

Based on Exam Data from December 1, 2025

1 Server Response Time (Z-Test for Mean)

We test the hypothesis about the population mean (μ) with known population standard deviation (σ).

1. Hypotheses

We are testing if the expected response time is equal to 400 ms.

$$H_0 : \mu = 400 \quad \text{vs} \quad H_A : \mu \neq 400 \quad (\text{Two-tailed test})$$

2. Given Data

$n = 20$, $\bar{X} = 394$, $\sigma = 10$, $\alpha = 0.05$.

3. Test Statistic (Z-statistic)

$$Z_{\text{obs}} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{394 - 400}{10/\sqrt{20}} \approx \frac{-6}{2.236} \approx -2.683$$

4. Critical Region and Decision

For a two-tailed Z-test at $\alpha = 0.05$, the critical value is $z_{0.025} = 1.96$. Since $|Z_{\text{obs}}| = 2.683 > 1.96$, the test statistic falls into the critical region. We **reject** H_0 at the 5% level of significance. We conclude that the expectation of the response time is significantly different from 400 ms.

2 Earthquake Strength (T-Interval for Mean)

We construct a 95% confidence interval for the population mean (μ) with unknown population standard deviation (s).

1. Given Data and Sample Statistics

- Data: 4.8, 3.9, 3.2, 2.8, 4.1.
- Sample size: $n = 5$. Degrees of freedom: $\nu = 4$.
- Sample mean: $\bar{X} = \frac{18.8}{5} = 3.76$.
- Sample standard deviation: $s \approx 0.8677$.

2. Critical Value and Confidence Interval

The critical t -value for 95% confidence and 4 d.f. is $t_{4;0.025} = 2.776$. The confidence interval is $\bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$:

$$3.76 \pm 2.776 \cdot \frac{0.8677}{\sqrt{5}} \approx 3.76 \pm 1.077$$

The 95% confidence interval for the expectation of the earthquake strength is:

$$[2.683, 4.837]$$

3 Moment Method Estimate for N

We estimate the unknown parameter N for a discrete uniform distribution on $\{N+1, N+2, \dots, 2N\}$.

1. Population Mean $E(X)$

For a discrete uniform distribution on $\{a, b\}$, the mean is $\frac{a+b}{2}$. Here, $a = N+1$ and $b = 2N$.

$$E(X) = \frac{(N+1) + 2N}{2} = \frac{3N+1}{2}$$

2. Moment Method Estimator ($\hat{N}$)

Equating the population mean to the sample mean ($\bar{X}$):

$$\bar{X} = \frac{3N+1}{2}$$

Solving for N gives the Moment Method Estimator:

$$\hat{N} = \frac{2\bar{X} - 1}{3}$$

4 Webshop Income (Linear Regression)

The model is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, where x is the month index (January=1, ..., May=5).

(a) Goodness of Fit

The goodness of fit is assessed by the R^2 value.

- $R^2 = 0.846$.

Since 84.6% of the total variation in income is explained by the linear model, the model shows a **very good fit** to the data.

(b) Significance of the Main Coefficient ($\hat{\beta}_1$)

We test $H_0 : \beta_1 = 0$ (no significant linear relationship) vs $H_A : \beta_1 \neq 0$ ($\alpha = 0.05$).

- P -value for the slope (month index) is 0.027.

Since $P = 0.027 < \alpha = 0.05$, we **reject** H_0 . The main coefficient (slope) is **significantly different from zero**, indicating a statistically significant linear trend in the income.

(c) Prediction for Income in August

August corresponds to $x = 8$. The estimated coefficients are:

- Intercept ($\hat{\beta}_0$): 4.1600
- Slope ($\hat{\beta}_1$): 0.4000

The prediction for August ($x = 8$) is:

$$\hat{y}_{\text{Aug}} = 4.1600 + 0.4000(8) = 4.1600 + 3.2000 = 7.36 \text{ million HUF}$$

5 Correlation Test (Z-Test for Proportions)

We test for a significant positive correlation (dependence) between speaking English (E) and German (G) ($\alpha = 0.05$).

1. Given Counts

$n = 200$, $n_E = 123$, $n_G = 86$ $n_{E \cap G} = 62$.

2. Hypotheses (One-Tailed Test)

$$H_0 : P(E \cap G) = P(E)P(G) \quad (\text{Independent})$$

$$H_A : P(E \cap G) > P(E)P(G) \quad (\text{Positively Dependent})$$

3. Test Statistic (Z-statistic Approximation)

The expected proportion under H_0 is $P_0 = \hat{P}(E)\hat{P}(G)$:

$$P_0 = \frac{123}{200} \cdot \frac{86}{200} \approx 0.615 \cdot 0.43 \approx 0.26445$$

The observed proportion is $\hat{P}_{\text{obs}} = \frac{62}{200} = 0.31$.

$$Z_{\text{obs}} \approx \frac{\hat{P}_{\text{obs}} - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} \approx \frac{0.31 - 0.26445}{\sqrt{\frac{0.26445(1-0.26445)}{200}}} \approx 1.463$$

4. Critical Value and Decision

For a one-tailed Z -test at $\alpha = 0.05$, the critical value is $z_{0.05} = 1.645$. Since $Z_{\text{obs}} \approx 1.463 < 1.645$, the test statistic does not fall into the critical region. We **fail to reject** H_0 . We conclude that there is **not** sufficient evidence to claim a significant positive correlation.